

Sparse Reconstruction based on K-Sparse Dictionary Learning Method for Bioluminescence Tomography

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Abstract— Bioluminescence tomography (BLT) is an important branch of molecular imaging, holds great prospect in biomedical research. However, BLT reconstruction results are often sensitive and inaccurate due to its inherently ill-posed property of inverse problem. Conventional BLT reconstruction methods based on dictionary learning suffer from slow convergence speed and prone to trap in sub-optimal solutions. To address these problems, this paper proposes an accelerated forward-backward splitting and the difference of convex functions algorithm (AFBS-DCA) based on dictionary learning framework. In the sparse coding stage, the non-convex Generalized Minimax-Concave (GMC) regularization is utilized to improve the sparsity of the sparse coefficient vector, the Nesterov's acceleration technique is integrated in AFBS algorithm to improve the global convergence performance and the k-sparsity strategy is innovatively introduced to achieve adaptive adjustment of the regularization parameter. In the dictionary updating stage, DCA is used to reduce the complexity of the objective function by decomposing the problem into easy and solvable sub-problems. Numerical simulation experiments verified the performance of the proposed method by comparing with two traditional reconstruction methods respectively. The results demonstrate that AFBS-DCA method has superior performance in terms of localization accuracy and shape recovery ability.

Clinical Relevance— The AFBS-DCA method proposed in this paper enhances BLT reconstruction efficiency and accuracy, providing new opportunities to deepen insights into disease mechanisms and assess drug efficacy in preclinical and clinical studies.

I. INTRODUCTION

Bioluminescence tomography (BLT) as a non-invasive and highly sensitive optical molecular imaging technique, has become a crucial tool in biomedical research [1]. It uses light propagation model and inversion algorithms to reconstruct the 3D distribution information of bioluminescent source in vivo based on the luminous flux measured on the surface of organisms [2].

In recent years, BLT technology has been widely used in the fields of early disease diagnosis, new drug development and treatment monitoring mainly because of its high sensitivity and non-invasive characteristics [3]. However, the reconstruction of BLT is faced with many challenges, such as the complex light scattering and absorption effects of biological tissues and the limitation of photon detection at the

boundary, etc. These factors seriously affect the accuracy and stability of reconstruction [4].

To solve the BLT reconstruction problem, researchers have proposed various methods, which can be categorized into two main classes: mathematical iteration methods and deep learning-based methods [5]. Deep learning-based approaches offer certain advantages such as remarkable flexibility, high accuracy, and rapid processing speed. For instance, the method proposed by Rezaeifar et al. demonstrates excellent performance in tumor targeting of BLT studies [6]. However, one of the primary challenges is the availability and quality of datasets. On the other hand, regularization plays a crucial role in mathematical iteration methods, particularly in the construction of optimization functions. Common loss functions include convex regularization such as L_1 -norm [7] and L_2 -norm [8], and non-convex sparse regularization (NSR) techniques. Algorithms based on L_1 -norm are widely used to address ill-posed problems, but it may lead to overly sparse reconstruction results, thereby adversely affects the accuracy of light source shape estimation [9]. Besides, regularization can also be integrated with Bayesian approaches. For instance, methods like the Gaussian Markov random field model, can preserve target edges and improve reconstruction accuracy [10]. Nevertheless, Bayesian methods often suffer from computational complexity, and their reliance on complex prior information limits their flexibility.

Apart from the methods previously mentioned, regularization can also be combined with dictionary learning, which is superior to Bayesian methods because it allows for more flexible and efficient sparse representation of data. dictionary learning alternates between sparse coding and dictionary update stages, optimizing both the dictionary and sparse representations iteratively [11]. For example, k-means singular value decomposition (K-SVD) [12] is a commonly used dictionary learning algorithm that iteratively optimizes the dictionary and sparse coefficients to minimize reconstruction error, but it has a high computational complexity, especially when the dictionary size is large.

Therefore, the accelerated forward-backward splitting and the difference of convex functions algorithm (AFBS-DCA) method based on dictionary learning framework is proposed in this paper, which adopts the update mechanism of alternating iteration optimization, including sparse coding and dictionary update. In the sparse coding stage, the NSR methods can strike

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a better balance between sparsity and accuracy than convex regularization techniques, the non-convex generalized minimax-concave (GMC) regularization [13] is utilized to improve the sparsity of the sparse coefficient vector, preserving the convexity of objective function and then we can solve it by using convex AFBS algorithm. In order to improve the convergence efficiency of the algorithm, the Nesterov's acceleration technique is incorporated to enhance both convergence and computational performance of algorithm. Furthermore, the k-sparsity strategy is introduced within the AFBS algorithm, achieving to adaptively adjust the regularization parameter, which eliminates the need for manual adjustment of regularization parameter and significantly improves the efficiency. In the dictionary update stage, DCA can decompose the objective function into a difference of convex functions to solve iteratively [14], effectively reducing the complexity of the objective function by decomposing the problem into differentiable and solvable sub-problems.

To evaluate the performance of the proposed method, we designed two sets of numerical simulation experiments. In these experiments, we compare the proposed algorithm with the incomplete variable truncation conjugate gradient algorithm (IVTCG) based on L_1 -norm [19] and the forward-backward splitting algorithm (FBS) [20].

II. METHOD

A. Forward Model

The organism has the characteristics of high scattering and low absorption, so the steady state diffusion equation (DE) with the Robin boundary condition can be utilized to describe the process of light propagation in the strictly controlled light-free environment, which is expressed as follows [15]:

$$\begin{cases} -\nabla \cdot [D(r)\nabla\phi(r)] + \mu_a(r)\phi(r) = S(r), r \in \Omega \\ \phi(r) + 2A(n)D(r)[\mathbf{v}(r) \cdot \nabla\phi(r)] = 0, r \in \partial\Omega \end{cases} \quad (1)$$

where S is the light intensity distribution of the internal light source; ϕ represents luminous flux density; Ω represents the whole area of the organism; $\partial\Omega$ is the boundary region of the organism; $\mathbf{v}$ is the outward normal vector; n is the refraction coefficient of the organism. $D(r) = 1/3[\mu_a + (1 - g)\mu_s]$ represents the diffusion coefficient at the position r , where μ_a and μ_s are absorption coefficient and optical scattering coefficient; g is the anisotropic parameter.

According to the Robin boundary condition, the theoretical value of the light intensity density transmitted from the biological surface can be derived as follows:

$$\Gamma(r) = -D(r)[\mathbf{v}(r) \cdot \nabla\phi(r)] = \phi(r)/2A(n) \quad (2)$$

Using the finite element method (FEM) to solve the photon transmission model, we can derive an important linear relational expression, which connects the distribution of light intensity density on the surface of the organism with the distribution of unknown spontaneous light sources inside [15]:

$$\mathbf{A}\mathbf{X} = \Phi \quad (3)$$

where $\mathbf{A}$ is a $m \times n$ system matrix, which contains the specific optical property information of each part of the organism. The light intensity density on the surface of the organism Φ can be obtained by practical measurement, and it

is a column vector of $m \times 1$ dimension. $\mathbf{X}$ is a coefficient vector with $n \times 1$ dimension.

BLT reconstruction is essentially ill-posed, so it is almost impossible to obtain a unique solution by solving Eq. (3) directly. To alleviate the ill-posedness of the inverse problem, the non-convex GMC regularization is employed as a penalty function to perform constraint optimization on Eq. (3), resulting in Eq. (4) [16]:

$$\min_{\mathbf{X}} \{F(\mathbf{X}) = \frac{1}{2} \|\mathbf{A}\mathbf{X} - \Phi\|_F^2 + \alpha J_{GMC}(\mathbf{X})\} \quad (4)$$

GMC regularization can preserve convexity of the cost function [13] [17], thereby avoiding false local minima in cost function, and achieving better reconstruction accuracy. The GMC punishment is shown as follows:

$$J_{GMC}(\mathbf{X}) = \|\mathbf{X}\|_1 - \min_{\mathbf{Z}} \{\|\mathbf{Z}\|_1 + \frac{\lambda}{2\alpha} \|\mathbf{A}(\mathbf{X} - \mathbf{Z})\|_F^2\} \quad (5)$$

where $\mathbf{Z}$ is auxiliary variable with $n \times 1$ dimension.

Based on the above GMC sparse regularization, we can construct the following dictionary learning cost function:

$$F(\mathbf{X}, \mathbf{A}, \mathbf{Z}) = \frac{1}{2} \|\mathbf{A}\mathbf{X} - \Phi\|_F^2 + \alpha \|\mathbf{X}\|_1 - \alpha \|\mathbf{Z}\|_1 - \frac{\lambda}{2} \|\mathbf{A}(\mathbf{X} - \mathbf{Z})\|_F^2 \quad (6)$$

In this work, we apply an alternative optimization scheme based on dictionary learning framework to solve the dictionary matrix $\mathbf{A}$, the coefficient vector $\mathbf{X}$ and the auxiliary variable $\mathbf{Z}$. The dictionary learning framework involves two stages: sparse coding and dictionary update, which are described in detail in the following.

B. Sparse Coding based on AFBS Algorithm

In the sparse coding stage, the GMC regularization preserves the convexity of the cost function, making it solvable by the AFBS convex optimization algorithm [17]. In the AFBS algorithm, we iteratively update objective coefficient vector $\mathbf{X}$ and auxiliary variable $\mathbf{Z}$ with a fixed variable $\mathbf{A}$, as follows:

$$(\mathbf{X}^{(k+1)}, \mathbf{Z}^{(k+1)}) = \argmin_{\mathbf{X}} \max_{\mathbf{Z}} F(\mathbf{X}, \mathbf{Z}) \quad (7)$$

where $F(\mathbf{X}, \mathbf{Z})$ is a saddle-point function, which is convex in $\mathbf{X}$ and concave in $\mathbf{Z}$. Therefore, the proximal gradient algorithm AFBS can be employed to obtain the global minimum.

To accelerate convergence, Nesterov's acceleration method [18] was employed for the updates of $\mathbf{X}$ and $\mathbf{Z}$. In addition, a k-sparsity strategy was introduced to adaptively adjust the regularization parameter λ [17]. The updating rules of AFBS algorithm are as follows:

$$\begin{cases} \omega_1^{(i)} = \mathbf{X}^{(i)} - \mu \mathbf{A}^T (\mathbf{A}\mathbf{X}^{(i)} - \Phi) + \mu \lambda \mathbf{A}^T \mathbf{A} (\mathbf{X}^{(i)} - \mathbf{Z}^{(i)}) \\ \omega_2^{(i)} = \mathbf{Z}^{(i)} + \mu \lambda \mathbf{A}^T \mathbf{A} (\mathbf{X}^{(i)} - \mathbf{Z}^{(i)}) \\ T^{(i)} = \omega_{1[k]}^{(i)} \\ \omega_1^{(i+1)} = (1 - \tau) \omega_1^{(i)} + \tau \omega_1^{(i-1)} \\ \omega_2^{(i+1)} = (1 - \tau) \omega_2^{(i)} + \tau \omega_2^{(i-1)} \\ \mathbf{X}^{(i+1)} = \text{soft}(\omega_1^{(i+1)}; T^{(i)}) \\ \mathbf{Z}^{(i+1)} = \text{soft}(\omega_2^{(i+1)}; T^{(i)}) \end{cases} \quad (8)$$

where the convergence condition are $0 < \mu < \frac{2}{\max(1, \frac{\lambda}{1-\lambda})} \|\mathbf{A}\|_2$ and $0 < \lambda < 1$; $\omega_{1[k]}^{(i)}$ denotes the k -th largest coefficient in the vector $\omega_1^{(i)}$; $\tau = \frac{1-t^{(k)}}{t^{(k+1)}}$, where the iterative formula for variable t is $t^{(k+1)} = \frac{1 + \sqrt{(1+4(t^{(k)})^2)}}{2}$.

In the traditional AFBS algorithm, the regularization parameter λ mainly controls the threshold setting in the soft thresholding operation, which means that the parameter λ determines how many coefficients will be set to zero during each iteration and how many coefficients will be retained in the soft threshold processing [20]. Under the new k -sparsity strategy, although λ still appears as a fixed parameter in the equations, manual adjustment of λ is no longer required. Instead, at each iteration, the threshold $T^{(i)}$, which serves as the soft thresholding parameter, is adaptively set to the k -th largest term (sorted by absolute value) in the coefficient vector $\omega^{(i)}$, that is $T^{(i)} = \omega_{[k]}^{(i)}$. This adaptive thresholding approach enables the soft thresholding operation to retain only the k largest coefficients while setting all other remaining coefficients to zero [17].

Through the sparse coding stage, the iteratively updated $\mathbf{X}$ and $\mathbf{Z}$ is obtained. Subsequently, in the dictionary update stage, the iterative update of the dictionary matrix $\mathbf{A}$ is carried out by fixing $\mathbf{X}$ and $\mathbf{Z}$.

C. Dictionary Update based on DCA

In the dictionary updating stage, the cost function has the form of the difference of two convex functions, so it can be expressed as a Difference of Convex functions (DC) program. We can use DCA to solve each convex sub-problem iteratively.

$$\min\{F(\mathbf{A}) = G(\mathbf{A}) - H(\mathbf{A}) | \mathbf{A} \in \mathbb{R}^{m \times n}\} \quad (9)$$

We define the convex DC components G and H , as follows:

$$G(\mathbf{A}) = \frac{1}{2} \|\mathbf{A}\mathbf{X} - \Phi\|_F^2 \quad (10)$$

$$H(\mathbf{A}) = \frac{\lambda}{2} \|\mathbf{A}(\mathbf{X} - \mathbf{Z})\|_F^2 \quad (11)$$

Since $H(\mathbf{A})$ is clearly defined and differentiable, DCA method is utilized to linearize the sub-differential of $H(\mathbf{A})$ and optimize Eq. (9). In each iteration of k , the sub-differential of DC components is as follows:

$$\begin{cases} \tilde{\mathbf{A}}^{(k)} = \nabla H(\mathbf{A}^{(k)}) = \lambda \mathbf{A}^{(k)}(\mathbf{X} - \mathbf{Z})(\mathbf{X} - \mathbf{Z})^T \\ \nabla G(\mathbf{A}^{(k)}) = (\mathbf{A}^{(k)}\mathbf{X} - \Phi)\mathbf{X}^T \end{cases} \quad (12)$$

In $k + 1$ iterations, we linearize the $H(\mathbf{A})$ of the objective function, and then the optimization problem $\mathbf{A}^{(k+1)}$ can be updated by solving for the following functions:

$$\begin{aligned} \mathbf{A}^{(k+1)} &= \underset{\mathbf{A} \in \mathbb{R}^{m \times n}}{\operatorname{argmin}} \{G(\mathbf{A}) - \langle \tilde{\mathbf{A}}^{(k)}, \mathbf{A} \rangle\} \\ &= \underset{\mathbf{A} \in \mathbb{R}^{m \times n}}{\operatorname{argmin}} \left\{ \frac{1}{2} \|\mathbf{A}\mathbf{X} - \Phi\|_F^2 - \langle \tilde{\mathbf{A}}^{(k)}, \mathbf{A} \rangle \right\} \quad (13) \end{aligned}$$

where $\tilde{\mathbf{A}}^{(k)}$ denotes the sub-differential of H , $\langle \tilde{\mathbf{A}}^{(k)}, \mathbf{A} \rangle$ represents the Frobenius inner product between two matrices, which is defined as: $\langle \tilde{\mathbf{A}}^{(k)}, \mathbf{A} \rangle = \operatorname{tr}(\tilde{\mathbf{A}}^{(k)T} \mathbf{A})$, where

tr denotes the trace of matrix, which is the sum of the elements on the main diagonal of a square matrix.

There are many ways to solve Eq. (13), but different optimization strategies have significant effects on the effect of dictionary update. For example, the gradient descent method can directly solve Eq. (13), but it usually converges slowly and the effect is unsatisfactory. We choose to rebuild Eq. (13) into a program in DC form and apply the DCA algorithm to optimize it. In able to use DCA, we optimize $\mathbf{A}_i$ column by column and decompose Φ into two convex functions, where Φ_1 and Φ_2 are defined as follows:

$$\Phi_1(\mathbf{A}_{i:}) = \frac{1}{2} \mathbf{A}_{i:} \operatorname{diag}(\eta)(\mathbf{A}_{i:})^T - \langle \tilde{\mathbf{A}}_{i:}^{(k)}, (\mathbf{A}_{i:})^T \rangle \quad (14)$$

$$\Phi_2(\mathbf{A}_{i:}) = \frac{1}{2} \mathbf{A}_{i:} \operatorname{diag}(\eta)(\mathbf{A}_{i:})^T - \frac{1}{2} \|\Phi_{i:} - \mathbf{A}_{i:}\mathbf{X}\|_2^2 \quad (15)$$

where $\eta = \mathbf{1}_{1 \times n} |M|$, $\operatorname{diag}(\eta)$ is the diagonal matrix constructed from the diagonal elements of $\mathbf{X}\mathbf{X}^T$ and used to ensure the convexity of decomposition.

Applying DCA, on the k -th iteration we can calculate $\tilde{\mathbf{Z}}^{(k)}$:

$$\tilde{\mathbf{Z}}^{(k)} = \nabla \Phi_2(\mathbf{A}_{i:}^{(k)}) = \mathbf{A}_{i:}^{(k)} \operatorname{diag}(\eta) - (\mathbf{A}_{i:}\mathbf{X} - \Phi_{i:})\mathbf{X}^T \quad (16)$$

On the $k + 1$ iteration, $\mathbf{A}^{(k+1)}$ can be calculated by solving for the following convex function:

$$\begin{aligned} \mathbf{A}_{i:}^{(k+1)} &= \underset{\mathbf{A}_{i:} \in \mathbb{R}^{m \times n}}{\operatorname{argmin}} \{ \Phi_1(\mathbf{A}_{i:}) - \langle \tilde{\mathbf{Z}}^{(k)T}, \mathbf{A}_{i:}^T \rangle \} \\ &= \{ \mathbf{A}_{i:}^{(k)} - \frac{[\nabla G(\mathbf{A}_{i:}^{(k)}) - \nabla H(\mathbf{A}_{i:}^{(k)})]}{[\eta]} \} \quad (17) \end{aligned}$$

At last, we need to normalize the columns of $\mathbf{A}$ to L_2 -norm. Therefore, the update of the dictionary matrix $\mathbf{A}$ is completed.

In summary, the AFBS-DCA algorithm with k -sparsity strategy is described in Algorithm 1.

Algorithm 1 AFBS-DCA algorithm for BLT reconstruction

Input: the system matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$, the photon measurement $\Phi \in \mathbb{R}^{m \times 1}$.

Parameter initialization: the parameter λ is positive constant, maximum parameter $\maxIter$, $iter = 0$, $t^{(1)} = 1$, $err = 1e - 6$, $\mathbf{X}^{(0)} \in \mathbb{R}^{n \times 1}$, and $\mathbf{A}^{(0)} \in \mathbb{R}^{m \times n}$ with each column normalized as unit 1.

While $\|\mathbf{X}^{(i)} - \mathbf{X}^{(i-1)}\|_2 > err$ or $iter < \maxIter$ **do**

Step1: Sparse coding updating rules based on Eq. (8)

Step2: Dictionary update flow

$$\nabla H(\mathbf{A}^{(j)}) = \lambda \mathbf{A}^{(j)}(\mathbf{X} - \mathbf{Z})(\mathbf{X} - \mathbf{Z})^T$$

$$\nabla G(\mathbf{A}^{(j)}) = (\mathbf{A}^{(j)}\mathbf{X} - \Phi)\mathbf{X}^T$$

$$\mathbf{A}_{i:}^{(j+1)} = \mathbf{A}_{i:}^{(j)} - \frac{[\nabla G(\mathbf{A}_{i:}^{(j)}) - \nabla H(\mathbf{A}_{i:}^{(j)})]}{[\eta]}$$

Step3: $iter = iter + 1$

End while

Output: $\mathbf{X} = \mathbf{X}^{(i+1)}$

III. EXPERIMENTS AND RESULTS

In this part, we designed two types of numerical simulation experiments to evaluate the effectiveness of AFBS-DCA algorithm for BLT reconstruction by comparing with IVTCG and FBS methods in terms of morphological recovery and positioning accuracy.

A. Numerical Simulation Experiment Setting

In the numerical simulation experiment, we constructed a model of a non-uniform cylinder of a specific size to simulate the structure of an organism with a radius of 10 mm and a height of 30 mm. The model contains five key organs, namely heart, bone, liver, lung, and muscle. Fig. 1(a) shows the geometric model used for numerical simulation in this experiment. We set specific optical parameters for each organ at a wavelength of 650 nm, the specific values of these parameters are shown in TABLE I. In this study, in order to carry out accurate numerical calculations of the established model, we used the COMSOL Multiphysics simulation software platform [21] to carry out discrete processing of the physical model and segmented it into a large number of uniformly distributed tetrahedral units. A grid system of 4626 nodes and 25840 tetrahedral elements was generated, and the mean element edge size is 1.37mm, as shown in Fig. 1(b) for the uniform tetrahedral mesh used in the reconstruction.

In the single light source experiment, we used a spherical light source with a radius of 1 mm and its center point set at the position of (-2, 4, 6) mm to simulate a real bioluminescent light source. Meanwhile, in the double light source experiment, we set up two such light sources, with their center points (-2, -8, 10) mm and (-2, -8, 14) mm respectively, to simulate two actual bioluminescent light sources.

TABLE I. OPTICAL PARAMETERS OF THE CYLINDRICAL MODEL.

Tissues	$\mu_a(\text{mm}^{-1})$	$\mu_s(\text{mm}^{-1})$	g
Muscle	0.0052	10.80	0.90
Heart	0.0083	6.733	0.85
Lung	0.0133	19.70	0.90
Liver	0.0329	7.000	0.90
Bone	0.0060	60.09	0.90

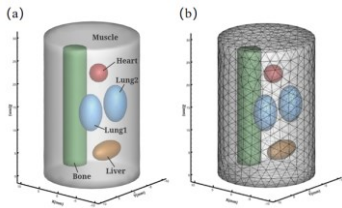


Figure 1. (a) Cylinder model for numerical simulation of single light source. (b) Uniform tetrahedral mesh for BLT reconstruction.

Furthermore, we introduced three pivotal quantitative evaluation indicators to evaluate the performance of these algorithms in terms of reconstruction quality and accuracy: localization error (LE), dice coefficient (DICE) [22], relative error (RE), and the contrast to noise ratio (CNR).

B. Single Light Source Numerical Simulation Results

Fig. 2 shows the reconstruction effect of the single light source simulation experiment, and the reconstruction results of three different algorithms are presented respectively. These images clearly reflect the performance of each algorithm in

light source position and shape recovery. The quantitative evaluation data of the three reconstruction methods are listed in detail in TABLE II.

From the quantitative results, we can conclude that AFBS-DCA method achieves the most accurate reconstruction result than other methods, with LE value of 0.312 mm and DICE value of 0.829. Compared to other methods, the reconstruction error is reduced by 46% and 48% respectively. The results demonstrate that AFBS-DCA algorithm has excellent performance in reducing positioning errors and improving the overall positioning accuracy of single light source reconstruction, and it can effectively restore the spatial distribution of real light sources.

TABLE II. QUANTITATIVE RESULTS OF SINGLE LIGHT SOURCE.

Method	LE (mm)	DICE	RE	CNR
IVTCG	0.674	0.307	0.937	0.261
FBS	0.648	0.648	0.593	0.423
AFBS-DCA	0.312	0.829	0.275	0.721

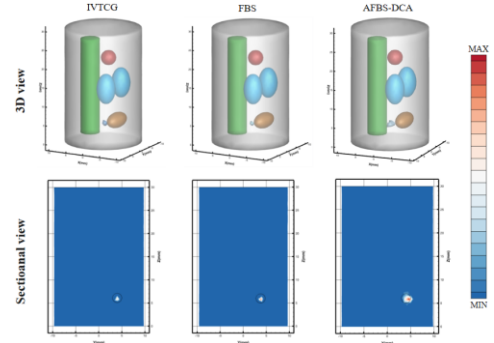


Figure 2. Comparison results of three algorithms in single light source numerical simulation experiment (3D view and sectional diagram).

C. Dual Light Source Numerical Simulation Results

Fig. 3 shows the reconstruction effect of the dual light source simulation experiment. It can be clearly seen by Fig. 3 that AFBS-DCA method shows a better performance in the accuracy of light source positioning and the coincidence of light source space compared with the other three methods, and its reconstruction effect is obviously best.

TABLE III. QUANTITATIVE RESULTS OF DUAL LIGHT SOURCE.

Method	AVG LE (mm)	AVG DICE	RE	CNR
IVTCG	1.217	0.585	1.137	0.111
FBS	0.690	0.555	0.678	0.283
AFBS-DCA	0.486	0.715	0.403	0.411

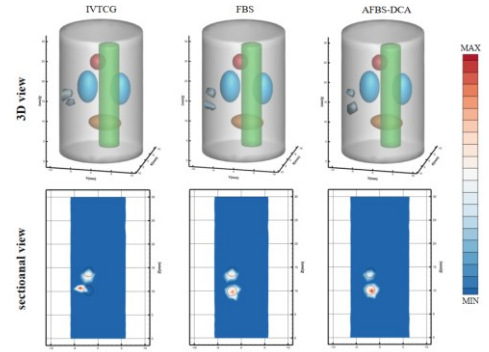


Figure 3. Comparison results of three algorithms in dual light source numerical simulation experiment (3D view and sectional diagram).

TABLE III. further supports the above inference from the perspective of quantitative analysis, which shows that the average LE value of AFBS-DCA method is 0.486 mm, the lowest among the three reconstruction methods, and the reconstruction error is reduced by 60% and 29% respectively. Meanwhile, the average DICE value of AFBS-DCA method is 0.715, which is the closest to 1.

IV. CONCLUSION

To improve the stability and accuracy of BLT reconstruction, this study proposes the AFBS-DCA method for BLT sparse reconstruction. The method is based on dictionary learning framework and employs an alternating iterative optimization strategy of sparse coding and dictionary updating. In the sparse coding stage, the AFBS algorithm combined with Nesterov's acceleration technique achieves faster and more stable reconstruction. What's more, we introduced a k-sparsity strategy to achieve the adaptive regularization parameter adjustment. In the dictionary update stage, the dictionary updating cost function with GMC regularization can be decomposed to the difference of two convex functions, so we can apply the DCA algorithm to solve the optimization problem. Numerical simulation experiments are designed to verify the effectiveness of AFBS-DCA method by comparing with IVTCG and FBS methods. The results demonstrate that AFBS-DCA method achieves more accurate reconstruction results in terms of morphological recovery and positioning accuracy than the other two methods. Moreover, the anti-noise experiment is conducted by adding varying noise levels to the input data. The results show that AFBS-DCA consistently achieves stable and high-quality reconstruction, demonstrating its robustness and adaptability.

Although AFBS-DCA has shown excellent performance in BLT reconstruction, it still has some limitations and shortcomings. Firstly, the AFBS-DCA method realizes the adaptive adjustment of parameters by introducing k-sparsity strategy, but some parameters still need to be adjusted manually, which is time-consuming. Secondly, the feasibility of AFBS-DCA method *in vivo* experiments has not been verified. Therefore, given the existing limitations of the AFBS-DCA method, future research will be devoted to solving the problem of manual parameter adjustment and verifying the feasibility of AFBS-DCA *in vivo*.

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